

Po-han Li

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RESEARCH INTERESTS

I study multimodal foundation models and human-aligned reasoning. I work on representation learning and cross-modal data translation—specifically video captioning, converting video into concise language and keyframe summaries, and turning sensor streams into structured symbolic traces—under data budgets and zero-GPU compute. I build lean evaluation frameworks for video-to-text generation that measure reliability and processing costs, expose failure patterns in captioning and summarization. I also create human-AI interaction systems that strengthen human oversight of VLMs and LLMs and improve alignment of their reasoning.

EDUCATION

University of Texas at Austin

Aug. 2023 – Present (Expected Summer 2026)

Ph.D. Candidate in Electrical and Computer Engineering

Texas, U.S.A.

- Decision, Information, and Communications Engineering (DICE) track
- Co-advised by Prof. [Sandeep Chinchali](#) and Prof. [Ufuk Topcu](#)
- GPA: 3.93/4.00

University of Texas at Austin

Aug. 2021 – May 2023

M.S. in Electrical and Computer Engineering

Texas, U.S.A.

- Decision, Information, and Communications Engineering (DICE) track

National Taiwan University

Sep. 2016 – Jul. 2020

B.S. in Electrical Engineering

Taipei, Taiwan

- Research Advisor: Prof. Wanjiun Liao (廖婉君)
- GPA: overall: 4.26/4.30 (3.99/4.0), last 60: 4.29/4.30. Ranking: 4/177
- Honors: Dean's List (2016 Fall, 2017 Spring, and 2018 Fall)
- College Student Research Scholarship from the Ministry of Science and Technology (2017-2019)

PUBLICATIONS

For a complete list of my publications, please check my [Google Scholar](#).

First-Author Publications:

1. N. P. Bhatt*, **P. Li***, and K. Gupta* et. al. UNCAP: Uncertainty-guided planning using natural language communication for cooperative autonomous vehicles. In *International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS) Best Paper Nomination*, 2026
2. **P. Li***, S. Chen*, S. Chinchali, and U. Topcu. VIBE: Video-to-text information bottleneck evaluation for TL;DR. *Advances in Neural Information Processing Systems (NeurIPS)*, 2025
3. **P. Li**, Y. Yang, M. Omama, S. Chinchali, and U. Topcu. Any2Any: Incomplete multimodal retrieval with conformal prediction. *Under review*, 2025
4. **P. Li**, S. Chinchali, and U. Topcu. CSA: Data-efficient mapping of unimodal features to multimodal features. *International Conference on Learning Representations (ICLR)*, 2025
5. **P. Li**, S. K. Ankireddy, R. Zhao, H. N. Mahjoub, E. Moradi-Pari, U. Topcu, S. Chinchali, and H. Kim. Task-aware distributed source coding under dynamic bandwidth. *Advances in Neural Information Processing Systems (NeurIPS)*, 2023

Other Publications:

1. M. Omama, **P. Li**, and S. Chinchali. Exploiting distribution constraints for scalable and efficient image retrieval. *International Conference on Learning Representations (ICLR)*, 2025

2. O. Akcin, **P. Li**, S. Agarwal, and S. Chinchali. Decentralized data collection for robotic fleet learning: A game-theoretic approach. In *Conference on Robot Learning (CoRL)*, 2022
3. Y. Geng, D. Zhang, **P. Li**, O. Akcin, A. Tang, and S. Chinchali. Decentralized sharing and valuation of fleet robotic data. In *Conference on Robot Learning (CoRL)*, 2021

WORK EXPERIENCE

- Meta Platforms, Inc.** May 2025 – Aug. 2025
Software Engineer Intern @ Monetization GenAI California, U.S.A
- Utilized automation tools to extract key information from external websites, supporting ML-driven advertisement generation
 - Aligned website source code with visual snapshot analysis, contributing to the design of evaluation pipelines for quality ad generation
 - Engineered robust evaluation metrics to assess the effectiveness of unsupervised generated advertisements
- Meta Platforms, Inc.** May 2024 – Aug. 2024
Software Engineer Intern @ Infra+Ranking & Foundational AI California, U.S.A
- Calibrated Meta's multimodal foundation AI model for ads ranking
 - Analyzed model performance and tracking of iterative training, supporting the development of robust evaluation benchmarks
 - Developed highly scalable classifiers and tools leveraging Python and machine learning techniques, incorporating data preprocessing and exploratory analysis
 - Adapted standard machine learning methods for distributed clusters, illustrating proficiency in model iteration and integrating heterogeneous data sources
- China Network Systems Co., Ltd.** Oct. 2019 – Mar. 2021
Machine Learning and Data Scientist Intern Taipei, Taiwan
- Analyzed data patterns and built prediction models for **churn rate (unsubscribe) prediction**
 - Used Raspberry Pi distributed in the core net and network terminals to collect network-quality data
 - Created databases and interactive reports to monitor **over 1M set-top boxes** in real-time
- Internet Research Lab** Aug. 2019 – Jun. 2020
Research Assistant @ National Taiwan University Taipei, Taiwan
- Participated in *5G mobile edge computing technology research and platform construction* project supported by the Ministry Of Science And Technology
 - Enhanced the quality of service (QoS) of multi-view 3D videos by **reinforcement learning**

ACADEMIC SERVICE

- Reviewer** Aug. 2021 - Present
- Reviewer of ICML, NeurIPS, ICLR, CVPR, MLSys, AAAI, AISTATS, IEEE Systems Journal, IROS, and ICRA.

EXTRACURRICULAR ACTIVITIES

- UT Girl Day**
- Volunteer Feb. 2023 and Feb. 2024
- Taught Python basics using AutoAuto cars; reached 200+ visitors of all ages. [link](#)
- Code2College**
- Mentor Jul. 2022 – Jul. 2024
- Mentored underrepresented high school graduates to prepare for software engineering jobs or college interviews.
- REACT REU**
- Mentor Aug. 2023
- Instructed undergraduate students to improve real-time computer vision models, excelling in image classification and object detection using Python.
- Student Council of National Taiwan University**
- Member of Parliament Jan. 2019 – May 2019
- Voiced concern about potential cyber security issues of the electrical voting system.